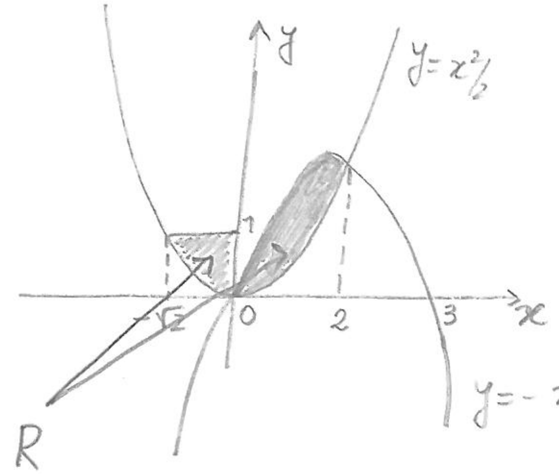
e)



$\begin{cases} y=-x^2+3x \\ y=\frac{x^2}{2} \end{cases}$ $\begin{cases} -x^2+3x=\frac{x^2}{2} \\ — \end{cases}$ $\begin{cases} 3x^2-6x=0 \\ — \end{cases}$ ②

$\begin{cases} 3x(x-2)=0 \\ — \end{cases}$ $\begin{cases} x=0 \lor x=2 \\ — \end{cases}$

$\frac{x^2}{2}=1 \Leftrightarrow x^2=2 \Leftrightarrow x=\pm\sqrt{2}$

$$\text{Área}(R)=\int_{-\sqrt{2}}^{0}\left(1-\frac{x^2}{2}\right)dx+\int_0^2\left((-x^2+3x)-\frac{x^2}{2}\right)dx$$

$$=\left[x-\frac{x^3}{6}\right]_{-\sqrt{2}}^{0}+\left[-\frac{\cancel{3}}{2}\frac{x^3}{\cancel{3}}+3\frac{x^2}{2}\right]_0^2$$

$$=\left(0-\left(-\sqrt{2}+\frac{2\sqrt{2}}{6}\right)\right)+\left(\left(-\frac{8}{2}+6\right)-0\right)$$

$$=\frac{2}{3}\sqrt{2}+2$$

② a) $f:]-1,1[\longrightarrow \mathbb{R}$, $t\longmapsto \frac{1}{1-t^4}$ é contínua e $\varphi:]-1,1[\longrightarrow\mathbb{R}$, $x\longmapsto x^2$ é derivável.

Então, pelo 1º Teorema Fundamental do Cálculo Integral, $F(x)=G\circ\varphi(x)$, em que $G(x)=\int_0^x f(t)\,dt$ é derivável e

$$F'(x)=G'(x^2)\,2x=\frac{2x}{1-x^8},\quad \forall x\in]-1,1[.$$

b) $$\frac{x}{1-x^4}+\frac{x}{1+x^4}=\frac{x(1+x^4)+x(1-x^4)}{(1-x^4)(1+x^4)}=\frac{2x}{1-x^8}$$

$$\int\frac{2x}{1-x^8}dx=\int\frac{x}{1-x^4}dx+\int\frac{x}{1+x^4}dx$$

$$=\frac{1}{2}\int\frac{2x}{1-(x^2)^2}dx+\frac{1}{2}\int\frac{2x}{1+(x^2)^2}dx$$

$$=\frac{1}{2}\operatorname{argth}(x^2)+\frac{1}{2}\operatorname{arctg}x+C,\ C\in\mathbb{R}$$

c) $$F(0)=\int_0^0\frac{1}{1-t^4}dt=0$$

Então

$\left.\begin{array}{l} F(x)=\frac{1}{2}\operatorname{argth}(x^2)+\frac{1}{2}\operatorname{arctg}x+C \\ F(0)=0 \end{array}\right\}\Rightarrow 0=\frac{1}{2}\operatorname{argth}(0)+\frac{1}{2}\operatorname{arctg}(0)+C$

$\Rightarrow C=0$

Então $$F(x)=\int_0^{x^2}\frac{1}{1-t^4}dt=\frac{1}{2}\operatorname{argth}(x^2)+\frac{1}{2}\operatorname{arctg}(x^2)$$

2º teste de Cálculo para Ciências

Correcção

1) a) $\int \operatorname{sen} x\, e^x dx = \operatorname{sen} x\, e^x - \int \cos x\, e^x = \operatorname{sen} x\, e^x - \left(\cos x\, e^x + \int \operatorname{sen} x\, e^x dx\right)$

CA1: $f = \operatorname{sen} x \quad f' = \cos x$
$g' = e^x \quad g = e^x$

CA2: $f = \cos x \quad f' = -\operatorname{sen} x$
$g' = e^x \quad g = e^x$

Então

$2\int \operatorname{sen} x\, e^x dx = \operatorname{sen} x\, e^x - \cos x\, e^x + C$

e

$\int \operatorname{sen} x\, e^x = \frac{1}{2} \operatorname{sen} x\, e^x - \frac{1}{2} \cos x\, e^x + C \quad , C \in \mathbb{R}$

b) $\int_0^1 \frac{x^2}{\sqrt{1-x^2}} dx = \int_0^{\pi/2} \frac{\operatorname{sen}^2 t}{\cancel{\cos t}} \cancel{\cos t}\, dt = \int_0^{\pi/2} \frac{1+\cos(2t)}{2} dt = \left[\frac{t}{2} + \frac{\operatorname{sen}(2t)}{4}\right]_0^{\pi/2}$

CA: $x = \operatorname{sen} t \quad dx = \cos t\, dt \quad x=0 \Rightarrow t=0$
$x = 1 \Rightarrow t = \pi/2$
$\sqrt{1-x^2} = \sqrt{1-\operatorname{sen}^2 t} = \cos t$

$= \left(\frac{\pi}{4} + \frac{\operatorname{sen}\pi}{4}\right) - (0+0) = \pi/4$

c) $\frac{x}{(x+1)(4x^2+4)} = \frac{A}{x+1} + \frac{Bx+C}{4(x^2+1)} = \frac{4A(x^2+1)+(Bx+C)(x+1)}{(x+1)(4x^2+4)}, \forall x \in \mathbb{R}$

$\Leftrightarrow \forall x \in \mathbb{R} \quad x = 4A(x^2+1) + (Bx+C)(x+1)$

$x=-1 \Rightarrow -1 = 8A \Leftrightarrow A = -1/8 \qquad x=0 \Rightarrow 0 = -1/2 + C \Leftrightarrow C = 1/2$

$x=1 \Rightarrow 1 = -1 + (B+1/2)2 \Leftrightarrow 1 = -1 + 2B + 1 \Leftrightarrow B = 1/2$

$\int \frac{x}{(x+1)(4x^2+1)} dx = -\frac{1}{8}\int \frac{dx}{x+1} + \frac{1}{8}\int \frac{x+1}{x^2+1} dx$

$= -\frac{1}{8} \ln|x+1| + \frac{1}{16}\int \frac{2x}{x^2+1} dx + \frac{1}{8}\int \frac{1}{x^2+1} dx$

$= -\frac{1}{8} \ln|x+1| + \frac{1}{16} \ln(x^2+1) + \frac{1}{8} \operatorname{arctg} x + C, \; C \in \mathbb{R}$

d) $\int_{-2}^{1} |(x-1)(x+1)| dx = \int_{-2}^{1} |x^2-1| dx = \int_{-2}^{-1} (x^2-1) dx + \int_{-1}^{1} (1-x^2) dx = ⊛$

C.A. $x^2 - 1 \geq 0 \Leftrightarrow x \in]-\infty, -1] \cup [1, +\infty[$

$⊛ = \left[\frac{x^3}{3} - x\right]_{-2}^{-1} + \left[x - \frac{x^3}{3}\right]_{-1}^{1} = \left[\left(-\frac{1}{3}+1\right) - \left(-\frac{8}{3}+2\right)\right] + \left[\left(1-\frac{1}{3}\right) - \left(-1+\frac{1}{3}\right)\right]$

$= \frac{2}{3} + \frac{2}{3} + \frac{2}{3} + \frac{2}{3} = \frac{8}{3}$